

Curriculum Vitae – HUGO NOWACKI

WORK EXPERIENCE

- Astrophysics Postdoc

1.5 years (03/2025 – 08/2026)

OCA, Nice, France.

Principal Investigator of the « Exoplanet Host Star » science program of the Interferometric Survey of Stellar Parameters (ISSP) project. Responsible for the characterization and optimization of the SPICA-FT fringe tracker at the CHARA array observatory. Supervision: Denis Mourard.

- Astrophysics Ph.D.

3 years (12/2021 – 12/2024)

IPAG, Grenoble, France

“Contribution of the VLT/GRAVITY+ instrument to the study of interactions between young stars and their proto-planetary disk”. Supervision: Karine Perraut, Jean-Baptiste Le Bouquin.

- System engineer

1 year (11/2020 – 11/2021)

IPAG, Grenoble, France

System engineer responsible for the low frequency radar JuRa onboard the Juventas nanosatellite for the ESA/Hera mission. Supervision: Alain Hérique, Yves Rogez.

- Astrophysics internship

4 months (03/2020 – 07/2020)

IPAG, Grenoble, France

“Mapping the magnetic field of the T Tauri star S CrA N through spectropolarimetric analysis”. Supervision: Evelyne Alecian, Karine Perraut.

- Thermal control engineering internship

6 months (03/2019 – 09/2019)

IAS, Orsay, France

Designing the thermal control system for the hyperspectral imager MacrOmega onboard the JAXA/MMX mission. Supervision: Vincent Hamm.

- Solar physics internship

3 months (05/2018 – 08/2018)

University of Sydney, Australia

Magneto-hydro-dynamic models of solar active regions and observational analysis of flare-related photospheric phenomena. Supervision: Michael Wheatland.

EDUCATION

- Ph.D. in astrophysics, Université Grenoble Alpes, France

2021 – 2024

- Master's in astrophysics, Université Grenoble Alpes, France

2019 – 2020

- Master's in astronomical and space-based system engineering (OSAE), observatoire de Meudon, France

2018 – 2019

- Bachelor's in fundamental physics, Université Paris-Sud, France

2016 – 2018

CONTRIBUTIONS TO CONFERENCES AND WORKSHOPS

• Invited presentations

- **Annual meeting of the AG** (~100 people) September 2025, Görlitz, Germany
Invited splinter talk: "Star-disk interactions in young Sun-like stars probed with VLT/GRAVITY" for the Astronomische Gesellschaft.
- **Observatory of Geneva seminar** (~30 people) May 2024, Geneva, Switzerland
Invited seminar: "The innermost regions of young Sun-like stars"
- **PROMETHEE ANR workshop** (~20 people) November 2023, Lyon, France
Invited talk: "Spectropolarimetric analysis of S CrA N and constraints from interferometry for the study of protostars"

• Contributed presentations

- **Cool Stars 22** (>500 people) June 2024, San Diego, California
Contributed plenary talk: "Perspectives on magnetospheric accretion in Classical T Tauri Stars".
- **GRAVITY+, impact on star and planet formation** (50 people) June 2024, Grenoble, France
Member of the LOC + contributed talk: "Towards real-time magnetospheric accretion monitoring with VLT/GRAVITY+"
- **GRAVITY consortium workshop** (~20 people) November 2023, Heidelberg, Germany
Contributed talk: "Is S CrA N growing its radiative core ?
Contribution of spectropolarimetry to the study of YSOs"

• Posters

- **The inner disk of young stars** (~100 people) May 2023, Cargèse, France
Poster: "Inner regions of the strongly accreting T Tauri Star S CrA N"
- **SPIE telescopes + instrumentation** (>500 people) July 2022, Montreal, Canada
Poster: "Impact of local turbulence on high-order adaptive optics"
- **Cool Stars 21** (>500 people) July 2022, Toulouse, France
Poster: "Structure of the inner regions in the T Tauri Star S CrA N"

INSTRUMENTATION & DATA MANAGEMENT

- **GRAVITY+ adaptive optics (GPAO)** VLT/Chile
Responsible for the characterization of the GPAO's deformable mirrors and their mount. Led their qualification in Paranal and contributed to their installation.
- **SPICA-FT Fringe tracker** CHARA, California
Responsible for the characterization and optimization of the SPICA-FT fringe tracker. Making phase-delay tracking routinely available for observations with the CHARA array 6-telescopes beam-combiners.
- **SPICA pipeline** CHARA, California
Contributed to the data reduction pipeline of CHARA/SPICA : improved the computation of uncertainties in the calibration process of interferometric data.

OBSERVING EXPERIENCE

- **Observing proposals :**

- **PI of the « First multi-chromatic imaging of young stellar objects »** CHARA, California
Four full nights allocated for 2025B and 2026A, aiming at combining R, H and K band interferometric observations from the SPICA, MIRC-X and MYSTIC instruments (program ID=SP2)

- **Col of the SPICA GTO, PI of the « Exoplanet host stars » program** CHARA, California
25 nights allocated per year (2021-2026) to combine SPICA, MIRC-X and MYSTIC. The exoplanet host stars (S01) program aims at measuring the size of stars to constraint the density and composition of their surrounding planets.

- **Col of 3 proposals focusing on the magnetic field of class I protostars** CFHT, Hawaii
28.3 hours in total allocated with the SPIRou instrument (infrared spectropolarimeter) to study the magnetic field and derive a magnetic map of class I protostars (IDs=24BF012, 25AF027, 25BF007).

- **Observing nights :**

- **SPICA, MIRC-X and MYSTIC interferometers** CHARA, California
2 weeks on-site (October 2025), and ~300 hours of remote observations for the ISSP project in 2025.
Operation of the three instruments simultaneously.

- **GRAVITY interferometer** VLT, Chile
1-week on-site as visitor (February 2024), and ~20 remote nights for the GRAVITY YSO group (2022 – 2024).

- **Neo-Narval spectropolarimeter** TBL, France
1-week service observations (February 2023).

- **SOPHIE spectrometer** OHP, France
1-week observations for a master's project (December 2021).

OTHERS

TEACHINGS:

- Three different teachings over two years for Université Grenoble Alpes: Real analysis seminar (72h), thermodynamics and mechanics practical classes (24h each).

PUBLIC OUTREACH:

- Supervised about 10 astronomy-night events, combining popularization of astronomy and sky observations for all audiences (~50 people each).
- Realized 2 podcasts explaining my Ph.D. to the public (in French)

MENTORING ACTIVITIES:

- Supervision of an undergraduate intern (3 months internship).
- Co-supervision of an undergraduate intern (3 months), and four high school interns (1 week).
- Coach and player for a local handball team for 15 years